Assignment 9: Sentiment Analysis of Emma

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Overview

In this assignment, I use the **tidytext** framework and three sentiment lexicons (**nrc**, **afinn**, and **bing**) to explore positive and joyful words in *Emma* by Jane Austen. I first prepare and tokenize the text, then join it with each lexicon, count frequent sentiment words, and visualize the results with bar charts and word clouds.

SetUp and Libraries Loading

Step 1: Data wrangling

In this step I load the 'Jane Austen' dataset and prepare the data for analysis. I create a "linenumber" variable to track each line and a "chapter" variable by detecting chapter headings using a regular expression. Finally, I filter the dataset to keep only the book 'Emma'.

```
emma_lines <- austen_books() %>%
  group_by(book) %>%
  mutate(
    linenumber = row_number(),
    chapter = cumsum(str_detect(
      text,
      regex("^chapter [\\divxlc]", ignore_case = TRUE)
    ))
  ) %>%
  ungroup() %>%
  filter(book == "Emma")
```

Step 2: Tokenization

In this step I convert the text into a tidy format using "unnest_tokens()", where each word becomes a separate row. This format is required for sentiment analysis.

```
emma_words <- emma_lines %>%
  unnest_tokens(word, text)
```

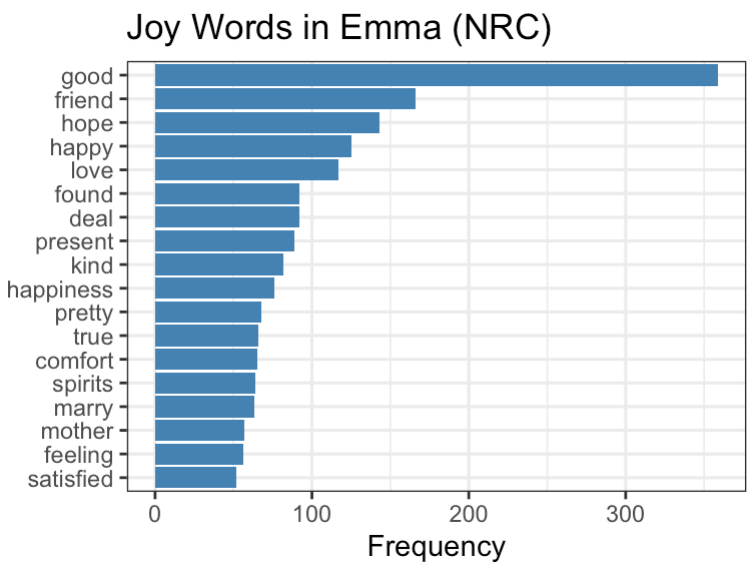
Step 3: NRC Sentiment Analysis (Joy)

Using the NRC lexicon I focus specifically on words associated with the emotion joy. I join the tokenized words with the NRC dataset, count word frequencies, and keep only the most frequent words (n>50).

```
emma_nrc <- emma_words %>%
  inner_join(
    get_sentiments("nrc") %>% filter(sentiment == "joy"),
    by = "word"
  ) %>%
  count(word, sort = TRUE) %>%
  filter(n > 50)
```

Visualization (Bar Chart)

```
emma_nrc %>%
  mutate(word = reorder(word, n)) %>%
  ggplot(aes(x = n, y = word)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Joy Words in Emma (NRC)",
    x = "Frequency",
    y = NULL
  )
```



Word Cloud

```
wordcloud(
  words = emma_nrc$word,
  freq = emma_nrc$n,
  min.freq = 1,
  max.words = 100,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)
```

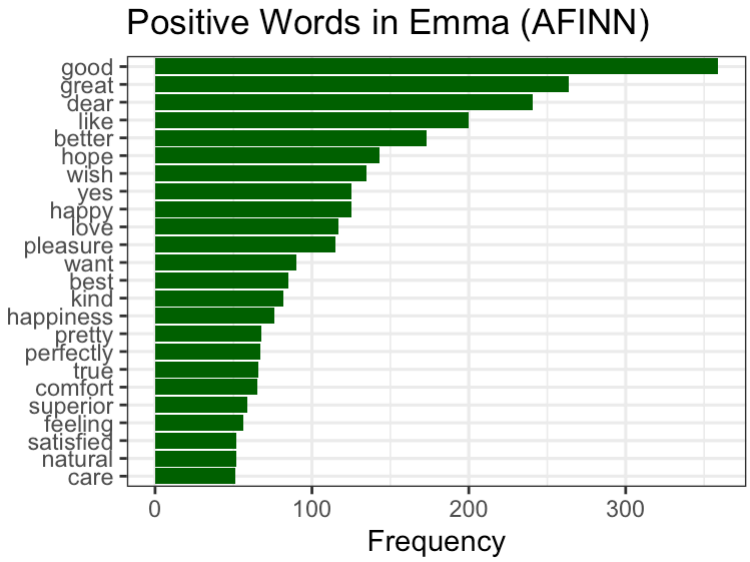
Step 3b: AFINN Sentiment Analysis (Positive)

The AFINN lexicon assigns numeric sentiment scores. In this step, I select only words with positive sentiment (Score>/= 1); then count and visualize the most frequent ones.

```
emma_afinn <- emma_words %>%
  inner_join(
    get_sentiments("afinn") %>% filter(value >= 1),
    by = "word"
  ) %>%
  count(word, sort = TRUE) %>%
  filter(n > 50)
```

Visualization (Bar Chart)

```
emma_afinn %>%
  mutate(word = reorder(word, n)) %>%
  ggplot(aes(x = n, y = word)) +
  geom_col(fill = "darkgreen") +
  labs(
    title = "Positive Words in Emma (AFINN)",
    x = "Frequency",
    y = NULL
    )
```



Word Cloud

```
wordcloud(
  words = emma_afinn$word,
  freq = emma_afinn$n,
  min.freq = 1,
  max.words = 100,
  random.order = FALSE,
  colors = brewer.pal(8, "Set2")
)
```

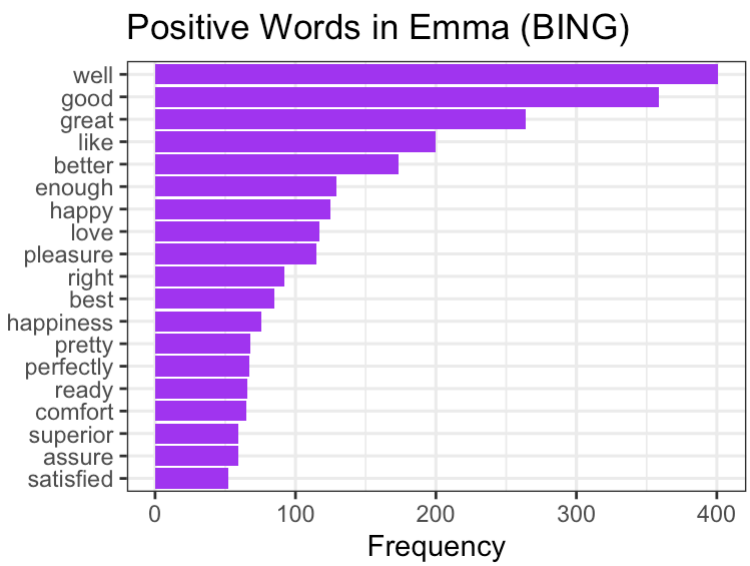
Step 3c: BING Sentiment Analysis (Positive)

The BING lexicon classifies words as either positive or negative. Here, I filter for positive words, count their frequency, and visualize the most common ones.

```
emma_bing <- emma_words %>%
  inner_join(
    get_sentiments("bing") %>% filter(sentiment == "positive"),
    by = "word"
  ) %>%
  count(word, sort = TRUE) %>%
  filter(n > 50)
```

Visualization (Bar Chart)

```
emma_bing %>%
  mutate(word = reorder(word, n)) %>%
  ggplot(aes(x = n, y = word)) +
  geom_col(fill = "purple") +
  labs(
    title = "Positive Words in Emma (BING)",
    x = "Frequency",
    y = NULL
  )
```



Word Cloud

```
wordcloud(
  words = emma_bing$word,
  freq = emma_bing$n,
  min.freq = 1,
  max.words = 100,
  random.order = FALSE,
  colors = brewer.pal(8, "Paired")
)
```

Conclusion

In this analysis I used three different sentiment lexicons to identify positive and joyful words in *Emma*. While each lexicon defines sentiment differently, all three highlight recurring positive language in the novel. The visualization show that *Emma* contains a consistent presence of emotionally positive vocabulary.